Graphic interface code

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="10dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">


    <TextView
        android:id="@+id/display"
        android:text="0"
        android:textSize="32sp"
        android:gravity="end"
        android:padding="20dp"
        android:background="#E0E0E0"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>


    <GridLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:columnCount="4">


        <Button android:text="C" android:onClick="onButtonClick"/>
        <Button android:text="+/-" android:onClick="onButtonClick"/>
        <Button android:text="%" android:onClick="onButtonClick"/>
        <Button android:text="/" android:onClick="onButtonClick"/>


        <Button android:text="7" android:onClick="onButtonClick"/>
        <Button android:text="8" android:onClick="onButtonClick"/>
        <Button android:text="9" android:onClick="onButtonClick"/>
        <Button android:text="*" android:onClick="onButtonClick"/>


        <Button android:text="4" android:onClick="onButtonClick"/>
        <Button android:text="5" android:onClick="onButtonClick"/>
        <Button android:text="6" android:onClick="onButtonClick"/>
        <Button android:text="-" android:onClick="onButtonClick"/>


        <Button android:text="1" android:onClick="onButtonClick"/>
        <Button android:text="2" android:onClick="onButtonClick"/>
        <Button android:text="3" android:onClick="onButtonClick"/>
        <Button android:text="+" android:onClick="onButtonClick"/>


        <Button android:text="0" android:onClick="onButtonClick"/>
        <Button android:text="." android:onClick="onButtonClick"/>
        <Button android:text="=" android:onClick="onButtonClick"/>
    </GridLayout>

</LinearLayout>
```

Logic code

```
package com.example.calculatorapp;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    TextView display;
    String current = "";
    String operator = "";
    double firstNumber = 0;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        display = findViewById(R.id.display);
    }

    public void onButtonClick(View view) {
        Button button = (Button) view;
        String text = button.getText().toString();

        switch (text) {

            case "C":
                current = "";
                operator = "";
                firstNumber = 0;
                display.setText("0");
                break;

            case "+/-":
                if (!current.isEmpty()) {
                    if (current.startsWith("-"))
                        current = current.substring(1);
                    else
                        current = "-" + current;

                    display.setText(current);
                }
                break;

            case "+":
            case "-":
            case "*":
            case "/":
                if (!current.isEmpty()) {
                    firstNumber = Double.parseDouble(current);
                    operator = text;


                    display.setText(current + " " + operator);
                    current = "";
                }
                break;

            case "=":
                if (!current.isEmpty() && !operator.isEmpty()) {
                    double secondNumber = Double.parseDouble(current);
```

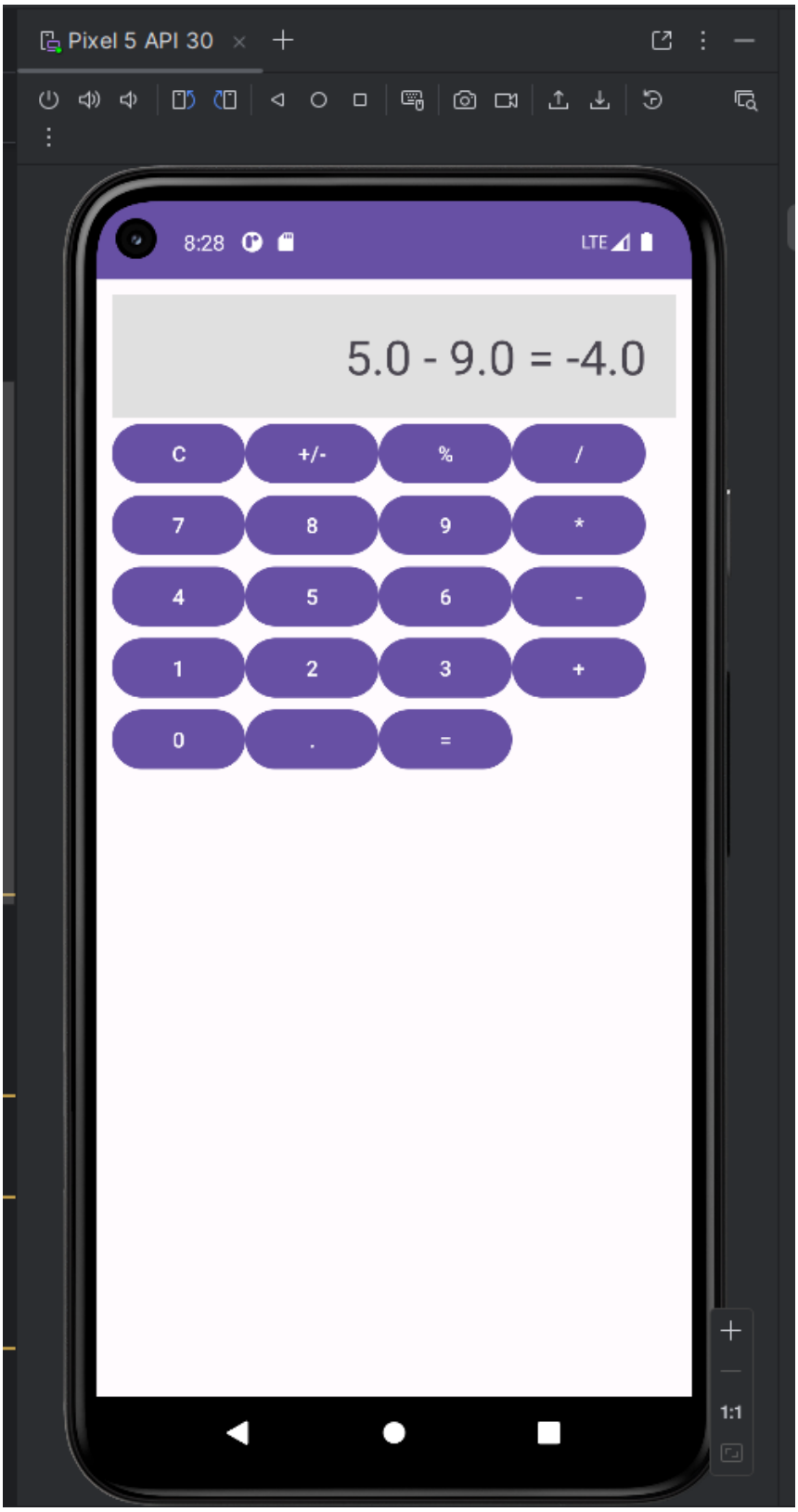
```
                    double result = 0;

                    switch (operator) {
                        case "+": result = firstNumber + secondNumber; break;
                        case "-": result = firstNumber - secondNumber; break;
                        case "*": result = firstNumber * secondNumber; break;
                        case "/":
                            if (secondNumber == 0) {
                                display.setText("Error");
                                return;
                            }
                            result = firstNumber / secondNumber;
                            break;
                    }


                    display.setText(firstNumber + " " + operator + " " + secondNumber +
" = " + result);

                    current = String.valueOf(result);
                    operator = "";
                }
                break;

            default:
                current += text;


                if (!operator.isEmpty()) {
                    display.setText(firstNumber + " " + operator + " " + current);
                } else {
                    display.setText(current);
                }
        }
    }
}
```

output